

ESTIMATION OF GROUNDWATER BALANCE IN THE PEDDAVANKA WATERSHED, ANANTAPUR DISTRICT, ANDHRA PRADESH.

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Abstract

An attempt has been made to estimate the groundwater balance in the Peddavanka watershed, Anantapur District, Andhra Pradesh. The annual groundwater recharge is measured from the average annual rainfall, recharge through surface water bodies and recharge through applied irrigation. The annual groundwater draft is estimated based on the number of wells present in the watershed and the extent of area irrigated. The draft based on the crop extent under different crops is also taken into consideration for both Kharif and Rabi seasons. From the study it is estimated that the annual groundwater recharge of Peddavanka watershed is 1995 ha.m. and the draft is estimated as 1615 ha.m. The annual groundwater balance of the Peddavanka watershed accounts for 380 ha.m which cannot be extracted due to the specific retention capacity of the soils present in the watershed.

Introduction

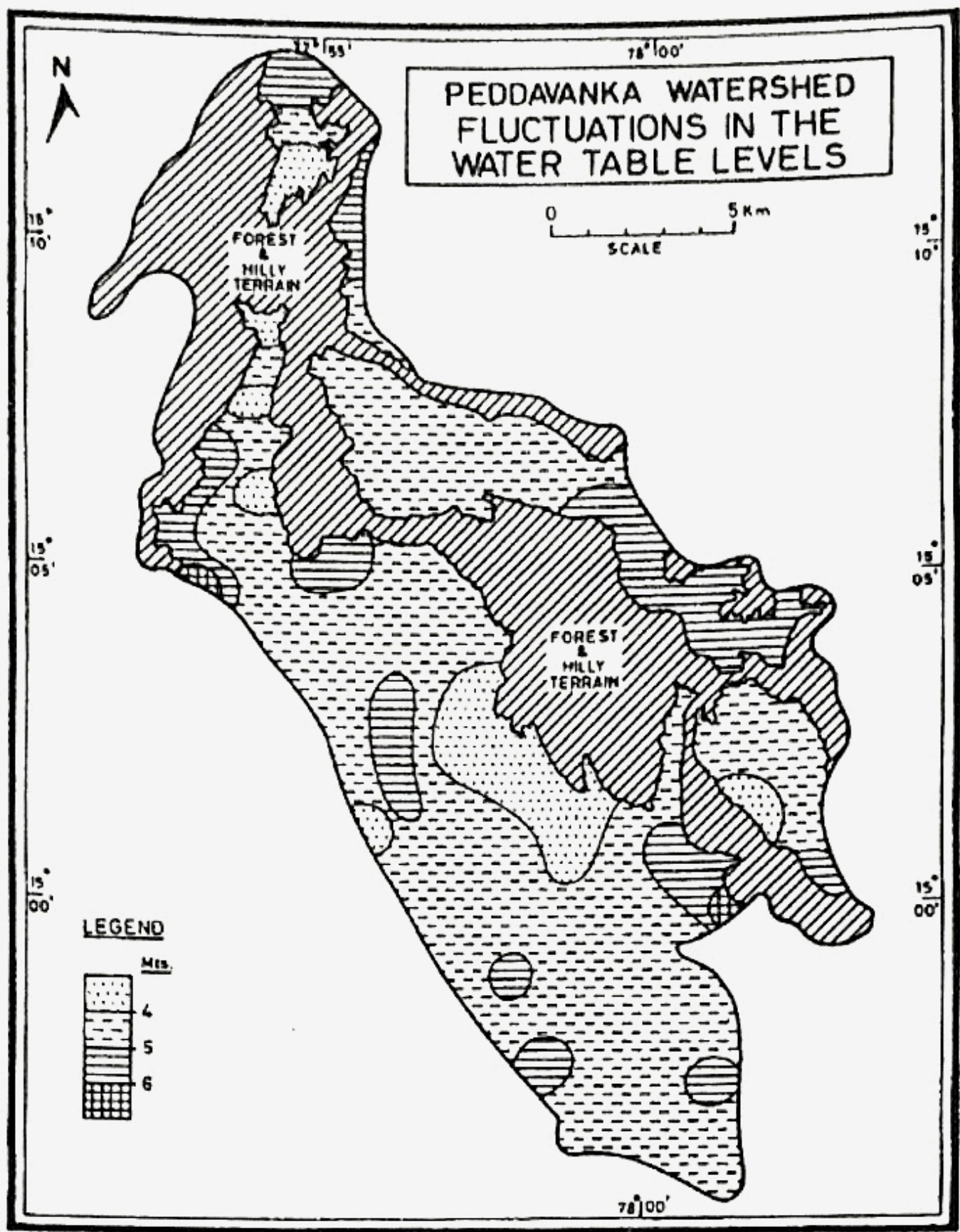
Estimation of groundwater balance of any area is important for the judicious management of the water resources. Peddavanka watershed is having a drainage area of 298 km² and showing dendritic to sub-dendritic drainage pattern. The watershed area lies between Latitudes 14°55'24" - 15°12'50" N and Longitudes 77°51'20" - 78°03'26" E. Seasonal water table fluctuation is a direct response of the groundwater draft and recharge in an area. Since rainfall is the principal source of groundwater recharge, water table rise has sympathetic relation to the rainfall, intensity and distribution of rainfall are the controlling factors on groundwater recharge. In addition, topography plays a vital role on the water table fluctuation and quantum of recharge. The annual fluctuation of water table is high in uplands and low in low lying/plains though the amount of rainfall is same in the watershed.

Climate

The watershed is influenced by semi-arid type of climate. It is observed that May is the hottest month of the year with a monthly mean temperature of 32.8° C and December is the coolest month with a monthly mean temperature of 23.7° C. The watershed area is deprived

The watershed is occupied by Tadipatri formation of Lower Cuddapah Supergroup which is mainly represented by shale with interbedded layers of quartzite and flaggy limestone. The shale trends NNE - SSW with about 10° dipping towards east. Shale often contains layers of basic volcanic rocks including sills of trap and trap conglomerate with volcanic pellets. Tadipatri formation is overlain by the rocks of the Kurnool group, which include Banaganapalle quartzite, Narji limestone and Auk shale. A distinct angular unconformity separates the cuddapah formation from the kurnool rocks, the latter occurring as horizontal beds over the tilted cuddapah rocks (Fig. 1).

water table level has direct relationship with rainfall in the watershed area. The monthly water table levels in two observation wells (Monitored by A.P.State Ground Water Department) for a period of four years (Jan., 1995 to Dec., 1998) are correlated with the monthly values of rainfall data. The groundwater levels show seasonal fluctuations influenced by rainfall and pumping of water from the wells for irrigation. In general waterlevel gradually increases during post-monsoon period (i.e., from July to December) and decrease during pre-monsoon (i.e., from January to June) seasons. Changes in climate also produce waterlevel fluctuations in the wells. During post monsoon season the climate is cool and evaporation is less while during pre-monsoon season the climate is hot and evaporation losses are maximum (Fig.4).



recommended by Groundwater Estimation Committee (GEC), CGWB, India and Groundwater Estimation Methodology (Government of India, Ministry of Irrigation, 1984). The details are given in Table.1.

Table 1. Groundwater Resource Evaluation

I. Recharge

A. Groundwater recharge by rainfall

1. Total geographical area of the basin	29,800 ha.
2. Area unsuitable for recharge occupied by hills and partly forests.	10,294 ha.
3. Area considered suitable for recharge	19,506 ha
4. Representative IMD normal annual rainfall (in mm)	515 mm
5. Assumed infiltration factor	15%
6. Rainfall recharge	1507 ha.m.

B. Recharge through water bodies

1. Water spread by tanks / ponds	67 ha.m.
2. Average depth of water in the tank	1000 mm
3. Seepage factor	50%
4. Total recharge to the ground	33.5 ha.m.

C. Recharge through applied irrigation (see draft for details)

1. Quantum of water applied without monsoon rainfall	1456 ha.m.
35% of recharge of groundwater for Paddy and 30% for remaining crops is considered out of quantum of water applied for irrigation	454.4 ha.m.

Total Recharge = A + B + C = 1507 + 33.5 + 454.4 = 1994.9 ha.m.

say 1995 ha.m.

II. Draft :

A. Draft due to the pumping wells

Number of wells present in the study area
(all the wells are fitted with pumpsets)
Rate of pumping from each well
Total pumping from each well in a day
(5 hours per day as pumping time)
Total withdrawal of groundwater from the wells
(180 days as pumping days)

Recharge is determined by considering recharge from rainfall, surface water bodies and recharge through the applied irrigation. In rainfall infiltration method, the area covered by hills and part of the reserved forest are considered as unsuitable for recharge, since these areas are mostly covered by rock exposures. The remaining part of the study area is considered as suitable for recharge through water bodies. The depth of water in the tanks / ponds is considered as 1 m (1000 mm) as an average for the entire watershed after taking measurements of depth in the tanks. From water bodies the recharge is estimated to be 33.5 ha.m. Then 35% of the applied water for Paddy irrigation and 30% of the applied water for other crops are considered as recharge (Ministry of Agriculture, 1971 and Water Technology Centre, 1977). From the above data it is estimated that the recharge of the watershed is 1995 ha.m.

Water Technology Centre, 1977. Water Requirement and Irrigation Management during
India. Indian Agricultural Research Institute, Monograph, No.4, Indian Agriculture Series
Station.